Search-Designed Trellis Codes with Neural Decoding

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1 Introduction and Motivation

Modern communication systems deployed in industrial IoT environments encounter structured interference that standard AWGN channel models do not capture. Periodic sinusoidal interference from switching power supplies, variable-frequency motor drives, and other electromagnetic sources imposes a deterministic-yet-unknown additive component on the received signal. Classical minimum-distance decoders, designed for white Gaussian noise, are mismatched to this channel and suffer avoidable performance degradation.

This project investigates the *joint co-design* of a convolutional trellis code and a learned neural decoder, both optimized for the sinusoidal interference regime. The central thesis is:

> *"When channels are mismatched, structured, or computationally constrained, the best end-to-end system may be obtained by co-designing the trellis and a constrained learned decoder for the realistic impairment family."*

The primary objective is not to surpass state-of-the-art codes on clean AWGN. The goals are:

1. **Robustness:** Maintain reliable decoding across a wide range of interference-to-noise ratios.
2. **Graceful degradation:** Performance degrades smoothly as interference parameters move outside the training distribution.
3. **Compute efficiency:** All methods operate within 2.1×10^6 operations per block, enabling fair comparison.

The system is evaluated using block error rate (BLER) at a target of BLER $= 10^{-3}$, across SNR $\in [0, 10]$ dB and INR $\in [-5, 15]$ dB.

2 System Model

2.1 Code Parameters

The system uses BPSK modulation with constellation $x[t] \in \{-1, +1\}$. Each block encodes $K = 256$ information bits at rate $R = 1/2$, producing $N = 512$ coded symbols. Table 1 lists all fixed system parameters.

2.2 Trellis Encoder

Definition 2.1 (Trellis Finite-State Machine)**.** *A trellis encoder is the tuple $\mathcal{T} = (\mathcal{S}, \mathcal{I}, \mathcal{O}, \delta, \lambda)$, where $\mathcal{S} = \{0, \ldots, S-1\}$ is the state space, $\mathcal{I} = \{0, 1\}$ is the binary input alphabet, $\mathcal{O} = \{-1, +1\}^2$ is the output symbol pair, $\delta : \mathcal{S} \times \mathcal{I} \to \mathcal{S}$ is the next-state function, and $\lambda : \mathcal{S} \times \mathcal{I} \to \mathcal{O}$ is the output labelling.*

Starting at initial state $s_0 = 0$, for each information bit $u_t \in \{0, 1\}$:

$$s_{t+1} = \delta(s_t, u_t), \tag{1}$$

$$\mathbf{c}_t = \lambda(s_t, u_t) \in \{-1, +1\}^2. \tag{2}$$

The block is *terminated* by appending tail bits that return the encoder to state 0, establishing $s_N = 0$ and enabling decoding with known boundary conditions. For the

Table 1: Fixed System Parameters

Parameter	**Value**
Modulation	BPSK, $x[t] \in \{-1, +1\}$
Information bits	$K = 256$
Coded symbols	$N = 512$
Code rate	$R = 1/2$
Trellis states	$S = 64$
Trellis structure	Time-invariant, block-terminated
Computational budget	2.1×10^6 operations per block
SNR range	0–10 dB (focus: 5 dB)
INR range	−5–15 dB
BLER target	10^{-3}
MC trials (evaluation)	10^5
MC trials (search full)	10^4
MC trials (search proxy)	10^3

$K = 7$ code, termination requires $K - 1 = 6$ tail bits, producing $2 \times (K_{\text{info}} + 6) = 524$ coded bits total.

2.3 Structural Constraints on Valid Trellis Candidates

All candidate trellises in Phase 3 must satisfy:

1. **Non-catastrophic:** No cycle in the trellis produces zero output weight.
2. **Full connectivity:** Every state is reachable from $s_0 = 0$, and s_0 is reachable from every state.
3. **Minimum free distance:** $d_{\text{free}} \geq 8$, providing correction of up to $\lfloor (8-1)/2 \rfloor = 3$ bit errors.

3 Channel Model and Mathematical Formulation

3.1 Received Signal

At discrete time $t \in \{0, \ldots, N-1\}$:

$$r[t] = x[t] + n[t] + i[t], \tag{3}$$

where $n[t] \sim \mathcal{N}(0, \sigma^2)$ is i.i.d. AWGN and $i[t]$ is the sinusoidal interference.

3.2 SNR and Noise Variance

With unit symbol energy $E_s = 1$:

$$\text{SNR}_{\text{lin}} = 10^{\text{SNR}_{\text{dB}}/10}, \qquad \sigma^2 = \frac{1}{\text{SNR}_{\text{lin}}}. \tag{4}$$

3.3 Sinusoidal Interference

$$i[t] = A \cdot \sin\left(\frac{2\pi t}{P} + \varphi\right), \tag{5}$$

with parameters drawn independently per block:

$$P \sim \mathcal{U}_{\mathbb{Z}}\{8, \ldots, 32\}, \tag{6}$$
$$\varphi \sim \mathcal{U}[0, 2\pi). \tag{7}$$

Since $P \geq 8$ and $N = 512$, at least $\lfloor 512/8 \rfloor = 64$ complete cycles occur per block.

3.4 Amplitude–INR Relationship

A sinusoid $A\sin(\cdot)$ has average power $A^2/2$. The INR is therefore:

$$\mathrm{INR}_{\mathrm{lin}} = \frac{A^2/2}{\sigma^2} \implies A = \sqrt{2\,\mathrm{INR}_{\mathrm{lin}}\,\sigma^2} = \sqrt{\frac{2\,\mathrm{INR}_{\mathrm{lin}}}{\mathrm{SNR}_{\mathrm{lin}}}}, \tag{8}$$

where $\mathrm{INR}_{\mathrm{lin}} = 10^{\mathrm{INR}_{\mathrm{dB}}/10}$.

Remark 3.1. *At INR = SNR = 5 dB (both linear ≈ 3.16), the amplitude $A = \sqrt{2} \approx 1.41$ exceeds the BPSK symbol magnitude of 1. Interference therefore dominates the signal at this operating point, making the mismatched AWGN assumption highly inaccurate.*

3.5 Training Distribution for Neural Methods

All channel parameters are re-randomised per batch block:

$$\mathrm{SNR}_{\mathrm{dB}} \sim \mathcal{U}[0, 10], \quad \mathrm{INR}_{\mathrm{dB}} \sim \mathcal{U}[-5, 10], \quad P \sim \mathcal{U}_{\mathbb{Z}}\{8, \ldots, 32\}, \quad \varphi \sim \mathcal{U}[0, 2\pi). \tag{9}$$

Evaluation uses both in-distribution and shifted test sets (e.g. INR $> 10\,\mathrm{dB}$, $P \notin [8, 32]$) to assess generalisation.

4 Baseline Methods

4.1 B1: Mismatched Viterbi Decoder

The primary baseline uses the NASA $K = 7$ rate-1/2 convolutional code with generator polynomials $\mathbf{g}_1 = [1, 1, 1, 1, 0, 0, 1]$ and $\mathbf{g}_2 = [1, 0, 1, 1, 0, 1, 1]$ (octal: 171, 133), achieving $d_{\mathrm{free}} = 10$.

The Viterbi decoder applies squared Euclidean branch metrics under the assumption of pure AWGN, ignoring interference:

$$\gamma_t^{\mathrm{mm}}(s \to s') = -\frac{1}{2\sigma^2} \sum_{j=1}^{2} \left(r_t^{(j)} - c_t^{(j)}\right)^2. \tag{10}$$

The true metric under the actual channel (Eq. 3) differs by a bias term involving $i_t^{(j)}$, which is structured and time-varying. B1 ignores this bias entirely.

The implementation uses a vectorized reverse-trellis formulation: for each destination state, all incoming branches are evaluated simultaneously via NumPy broadcasting. An optional C extension (`viterbi_core.so`) provides additional speedup when compiled. The Viterbi algorithm performs add-compare-select operations at each of $T = N/2 = 256$ time steps across $S = 64$ states with 2 incoming branches per state.

4.2 B2: Oracle Viterbi Decoder (Performance Ceiling)

The oracle decoder knows A, P, φ exactly per block, subtracts $i[t]$, and applies the matched metric to the cleaned signal $\tilde{r}[t] = r[t] - i[t] = x[t] + n[t]$:

$$\gamma_t^{\text{oracle}}(s \to s') = -\frac{1}{2\sigma^2} \sum_{j=1}^{2} \left(\tilde{r}_t^{(j)} - c_t^{(j)} \right)^2 . \tag{11}$$

B2 represents the performance ceiling. The gap B1 − B2 (in dB at fixed BLER) quantifies the loss attributable solely to interference ignorance.

4.3 B3: Random Trellis + Neural Decoder (Search Sanity Check)

B3 uses a randomly generated 64-state trellis (subject to structural constraints) with the same GRU decoder as N1. Its sole purpose is to verify that Phase 3 search finds genuinely good structure: the searched trellis (S1) must beat B3 by $\geq 1\,\text{dB}$ at BLER $= 10^{-3}$. B3 is not yet evaluated; it is scheduled for Phase 3.

4.4 B5: Interference Estimation and Cancellation + Viterbi

B5 estimates (A, P, φ) from the received signal via DSP, cancels the estimated interference, then applies mismatched Viterbi to the residual:

1. **Frequency estimation.** Compute the FFT of the received signal and identify the dominant frequency within the valid period range $P \in [8, 32]$:

$$\hat{f} = \arg \max_{f' \,:\, 1/32 \leq f' \leq 1/8} \left| \sum_{t=0}^{N-1} r[t]\, e^{-j2\pi f' t} \right|^2 . \tag{12}$$

   The top-5 frequency candidates are retained for refinement.

2. **Amplitude and phase estimation.** For each candidate frequency $\hat{f}$, form the least-squares problem with basis $[\cos(\omega t),\ \sin(\omega t),\ 1]$ where $\omega = 2\pi \hat{f}$:

$$\min_{a,b,c} \sum_{t=0}^{N-1} \left(r[t] - a\cos(\omega t) - b\sin(\omega t) - c \right)^2 . \tag{13}$$

   The amplitude and phase are recovered as $\hat{A} = \sqrt{a^2 + b^2}$ and $\hat{\varphi} = \arctan(a/b)$. The candidate with the lowest residual sum-of-squares is selected.

3. **Cancellation and decoding.** $\tilde{r}[t] = r[t] - \hat{A}\sin(2\pi t/\hat{P} + \hat{\varphi})$, then B1 Viterbi on $\tilde{r}[t]$.

B5 is critical for determining the regime where learning adds value: if simple DSP cancellation closes the gap between B1 and B2, the neural approach offers no benefit.

5 GRU Neural Decoder (N1)

The GRU end-to-end decoder (N1) is scheduled for Phase 2a and has not yet been implemented.

6 Neural Branch Metric Estimator (N2)

The neural branch metric estimator (N2) is scheduled for Phase 2b and has not yet been implemented.

7 Interference Estimation Details

The interference estimation and cancellation module (`interference_est.py`) is fully implemented and integrated into the B5 baseline as described in Section 4.4.

8 Trellis Search Algorithm

The evolutionary trellis search is scheduled for Phase 3 and has not yet been implemented.

9 Experimental Results

9.1 Validation Tests

All ten validation tests pass as of March 11, 2026. Table 2 summarises the three categories of validation.

Table 2: Validation Test Results (pytest, 10 tests total)

Test	Description	Status
Noiseless round-trip (4 tests)	Encode → Viterbi decode in zero noise: 0 bit errors across 100 random blocks, all-zero block, output length, binary check	PASS
AWGN theory match (3 tests)	BLER at 8 dB < 0.05; BLER decreases with SNR; BLER at 0 dB > 0.5	PASS
Oracle dominance (3 tests)	Oracle BER $\leq$ mismatched BER at INR = 5 dB and 10 dB; identical output with zero interference	PASS

9.2 BLER vs. SNR Performance

The full Phase 1 Monte Carlo evaluation (`eval.py --n-trials 100000`) has not yet been executed. No `.npz` result files exist in `results/phase1/`. Accordingly, no BLER values are available to report.

9.3 BLER vs. INR — Robustness Analysis

10 Compute Budget Analysis

All methods must operate within 2.1×10^6 operations per block. Table 3 reports both analytical formulas and measured values from `compute_cost.py`.

The Viterbi decoder complexity is computed as $S \times 2 \times T \times n_{\text{outputs}}$ where $S = 64$, $T = N/n_{\text{outputs}} = 256$, and $n_{\text{outputs}} = 2$:

$$\text{FLOPs}_{\text{Viterbi}} = 64 \times 2 \times 256 \times 2 = 65{,}536. \tag{14}$$

[Figure pending: run `eval.py` to generate BLER vs. SNR curves for B1, B2, B5.]

Figure 1: BLER vs. SNR at INR = 5 dB. Monte Carlo trials: 10^5. Pending execution of `eval.py`.

[Figure pending: INR sweep at SNR = 5 dB.]

Figure 2: BLER vs. INR at SNR = 5 dB for all implemented methods. Pending execution.

The interference estimation cost includes an FFT contribution of $5N \log_2 N$ complex operations, a least-squares component of $9N$, and a cancellation step of $3N$:

$$\text{FLOPs}_{\text{IC}} = 5 \times 512 \times 9 + 9 \times 512 + 3 \times 512 = 23{,}040 + 4{,}608 + 1{,}536 = 29{,}184. \quad (15)$$

The total for B5 is $65{,}536 + 29{,}184 = 94{,}720$.

Table 3: Measured Computational Cost Per Block (from `compute_cost.py`)

Method	**Dominant operation**	**Measured FLOPs**	**% of Budget**
B1: Mismatched Viterbi	Add-compare-select	65,536	3.1%
B2: Oracle Viterbi	Add-compare-select	65,536	3.1%
B5: IC + Viterbi	FFT + ACS	94,720	4.5%
N1: GRU ($H = 64$)	Recurrent GEMM	[TBD — Phase 2a]	—
N2: Neural BM	Per-transition MLP	[TBD — Phase 2b]	—

Remark 10.1. *All three implemented baselines operate well within the* 2.1×10^6 *operation budget, using at most* 4.5% *of the budget. The* `compute_cost.py` *profiler confirms that* `assert_within_budget` *passes for B1, B2, and B5. This leaves substantial headroom for neural methods, though the analytical GRU estimate of* $N(4H + 4H^2 + H) \approx 8.55 \times 10^6$ *MACs exceeds the budget and will require architectural adjustment in Phase 2a.*

11 Ablation Study

Table 4 presents the planned ablation matrix. All methods will operate at the equal 2.1×10^6 operation budget. Results are populated during Phase 4.

Table 4: Ablation Matrix — BLER at SNR = 5 dB, INR = 5 dB, 10^5 MC trials

ID	Trellis	Decoder	Role	BLER
B1	K=7 classical	Mismatched Viterbi	Primary baseline	—
B2	K=7 classical	Oracle Viterbi	Performance ceiling	—
B5	K=7 classical	IC + Viterbi	DSP cancellation baseline	—
N1	K=7 classical	GRU end-to-end	Decoder contribution	—
N2	K=7 classical	Neural BM + Viterbi	Structured hybrid	—
B3	Random	GRU end-to-end	Search sanity check	—
S1	Searched	GRU end-to-end	**Core result**	—
S2	Searched	Neural BM	Optional extension	—

12 Discussion and Analysis

Discussion will be added once the Phase 1 Monte Carlo evaluation has been executed and BLER curves are available.

13 This Week's Progress

The following tasks were completed to bring Phase 1 to completion:

- Implemented `channel.py`: AWGN generation, sinusoidal interference with the amplitude formula $A = \sqrt{2 \cdot \mathrm{INR}_{\mathrm{lin}} \cdot \sigma^2}$, BPSK modulation/demodulation, full channel model, and channel parameter sampling. All smoke tests pass.
- Implemented `trellis.py`: trellis FSM dataclass with encode and validate methods, NASA K=7 code loader (generators 171, 133 octal), random trellis generator for baseline B3, and block termination. Validation confirms the NASA code is non-catastrophic, fully connected, and terminable.
- Implemented `decoders.py`: vectorized Viterbi decoder using a reverse-trellis formulation with NumPy broadcasting, optional C extension via ctypes, mismatched branch metric (Eq. 10), and oracle branch metric with interference subtraction.
- Implemented `interference_est.py`: FFT-based period estimation over the valid range $P \in [8, 32]$, least-squares amplitude/phase fitting with top-5 frequency candidates, and combined estimate-and-cancel pipeline.
- Implemented `baselines.py`: wrapper functions for B1, B2, and B5 that integrate the encoder, channel, decoders, and interference estimation modules.
- Implemented `compute_cost.py`: analytical FLOPs counting for Viterbi and IC+Viterbi, latency measurement via `time.perf_counter`, budget assertion, and markdown table generation. Measured values: B1/B2 at 65,536 FLOPs (3.1% of budget), B5 at 94,720 FLOPs (4.5% of budget).

- Implemented `eval.py`: Monte Carlo BLER estimation with Wilson confidence intervals, SNR sweep with incremental saving, method wrapper factories for B1/B2/B5, and Phase 1 evaluation runner with automatic plotting and compute-cost reporting.
- Implemented `plot_utils.py`: paper-quality BLER vs. SNR and BLER vs. INR plotting functions with consistent method styling, confidence interval bands, and dual PDF/PNG output.
- Wrote and passed all 10 validation tests across three test files: noiseless round-trip (4 tests), AWGN theory consistency (3 tests), and oracle-vs-mismatched dominance (3 tests).

Issues currently open:

- The full Phase 1 Monte Carlo evaluation (`eval.py` with 10^5 trials) has not yet been run. All code is ready; execution is pending to generate the BLER curves and populate the results directory.
- The analytical GRU operation count ($\approx$ 8.55M MACs for H=64) exceeds the 2.1M budget by $\sim$4$\times$. Architecture adjustments will be needed in Phase 2a.

14 Next Steps

The following tasks are scheduled for Phase 2a (GRU End-to-End Decoder), drawn from the Phase 2 checklist in `CLAUDE.md`:

1. `neural_decoder.py`: GRU with 64 hidden units, verify $\leq$ 2.1M ops via op counter.
2. Training loop: randomize SNR $\in [0, 10]$ dB, INR $\in [-5, 10]$ dB, $P \in [8, 32]$, $\varphi \in [0, 2\pi]$ per batch.
3. Evaluate: generalization to INR/P values outside training distribution.
4. Compare N1 vs B1 — must beat by ≥ 0.5 dB at BLER $= 10^{-3}$ for Phase 2 success.

The success criterion for Phase 2a is: N1 beats B1 by ≥ 0.5 dB at BLER $= 10^{-3}$; training is stable.